



MIAE 215 Final Exam Pointers Assignment Problems and Solutions

Course: [Mechanical, Industrial & Aerospace engineering \(MIAE 215\)](#) 378 documents

University: [Concordia University](#)

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MIAE 215 Pointers Assignment

Please work the assignment problems carefully by hand in a manner similar to a written exam before you look at the output/solutions. For instance, if you see a for-loop modifying an array then list the array elements after the loop. Don't try to do the problems in your head. Write down the new variable values when they are changed so you can keep track of them (this is also useful for part marks). These types of problems are typical pointer questions for a written exam.

Note: when cutting and pasting these problems into a C++ compilers the quotation marks can get messed up and thus need to be replaced.

1) What is the output of the program below ?

```
double *p1, *p2, i1=3, i2=4;
```

```
p1 = &i1;  
p2 = &i2;
```

```
*p1 = 2;  
i2 = 7;
```

```
cout << "\n" << i1;  
cout << "\n" << *p2;
```

```
p2 = p1;
```

```
i1 = 4;  
*p2 = 9;
```

```
cout << "\n" << i1;
```

2) What is the output of the program below ?

```
double *p, x[4]={1.1, 2.2, 3.3, 4.4};
```

```
p = x;
```

```
cout << "\n" << x[2];  
cout << "\n" << p[3];  
cout << "\n" << *(p+1);  
cout << "\n" << *p;
```

3) What is the output of the program below ?

```
double *p, x[4]={1.1, 2.2, 3.3, 4.4};
```

```
p = x - 1;
```

Mock exam

Summary

Quiz

Ask a question about this document





```
*p1 = 2;
i2 = 7;

cout << "\n" << i1;
cout << "\n" << *p2;
```

```
p2 = p1;
```

```
i1 = 4;
*p2 = 9;
```

```
cout << "\n" << i1;
```

2) What is the output of the program below ?

```
double *p, x[4]={1.1, 2.2, 3.3, 4.4};
```

```
p = x;
```

```
cout << "\n" << x[2];
cout << "\n" << p[3];
cout << "\n" << *(p+1);
cout << "\n" << *p;
```

3) What is the output of the program below ?

```
double *p, x[4]={1.1, 2.2, 3.3, 4.4};
```

```
p = x - 1;
```

```
cout << "\n" << x[2];
cout << "\n" << p[3];
cout << "\n" << *(p+4);
```

4) What is the output of the program below ?

```
int main()
{
    double *p1, *p2;
    int i, n;

    n = 10;
    p1 = new double [n];
    p2 = p1 + 2;

    for(i=0;i<(n-2);i++) {
        p1[i] = i;
        *(p2+i) = -i;
    }

    cout << "\n" << *(p1+2);
    cout << "\n" << p2[3];

    delete
    cout <
    return 0;
```

delete

Mock exam

Summary

Quiz

cout <

Ask a question about this document





```
cout << "\n" << x[2];
cout << "\n" << p[3];
cout << "\n" << *(p+4);
```

4) What is the output of the program below ?

```
int main()
{
    double *p1, *p2;
    int i, n;

    n = 10;
    p1 = new double [n];
    p2 = p1 + 2;

    for(i=0;i<(n-2);i++) {
        p1[i] = i;
        *(p2+i) = -i;
    }

    cout << "\n" << *(p1+2);
    cout << "\n" << p2[3];

    delete (p2-2);

    cout << "\n\ndone.\n";

    return 0;
}
```

5) What is the output of the program below ?

```
double fun1(double *a, int *b) {

    a[0] = 1.2;
    *b = 3;

    return a[1]*a[2];
}

void main()
{
    int i=7;
    double x[4]={1.1, 2.2, 3.3, 4.4};

    cout << "\n" << fun1(x,&i);
    cout << "\n" << i;
    cout << "\n" << x[0];
}
```

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